CS215: Data Analysis and Interpretation

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(Many slides obtained from Prof. Ajit and Prof. Sunita and gratefully acknowledged. Fair use of internet resources as available to professors under copyright act)

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Agenda

- Last time
  - Random variables
  - *Expectation* of a random variable
  - Expectation of a function of a random variable
- Today
  - More on random variables
  - Time permitting: Grading, Evaluation, Academic Honesty, Deadline extension, “what if I missed”, …

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Why should this work?

Control of the complicated pdf of a face using an artificial neural network

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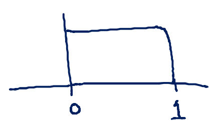
Uniform

Canonical

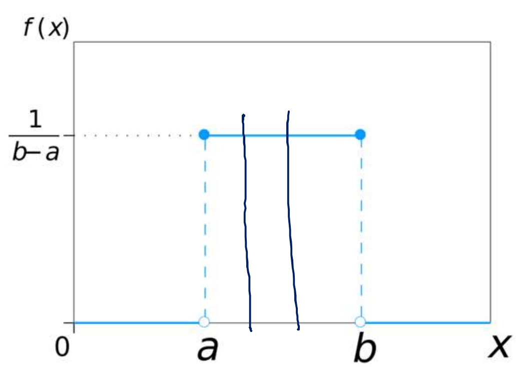


$p_X + p_Y = p(x+y)$

?

0.3
0.9

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b \\ 0, & \text{otherwise} \end{cases}$$



If X is uniform between 0 and 1, and Y is identical, what is the distribution of X+Y?

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Random variable: notation

- A random variable is denoted by an upper case letter.
- The individual values it can take are denoted by the matching lower case letter.
  - The probability that the random variable X takes the value x

$$P(X = x)$$

Expectation

For a **discrete** random variable *X*:

For a **continuous** random variable *X*:

Expected value of X is also called mean and represented by the symbol μ

- The expected value when you throw a die is 1/6 (1+2+3+4+5+6) = 3.5
- We will never get 3.5 when we throw a dice

$$E(X) = \sum_i x_i \, P(X = x_i)$$

$$E(X) = \int_{-\infty}^{+\infty} x \, f_X(x) \, dx$$

Function of a random variable

For a function *g*(*X*) of a **discrete** random variable $E(g(X)) = \sum_i g\,(x_i)P(X = x_i)$

For a **continuous** random variable $E(g(X)) = \int_{-\infty}^{+\infty} g\,(x) f_X(x)\, dx$

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Illustration (Discrete)

$X = 0, 1, 2$

$0.2 \quad 0.5 \quad 0.3$

$Y = X^2$

$0, 1, 4$

$P_y = 0.2 \quad 0.5 \quad 0.3$

$E[Y] = \sum g(x) P_{x_i} = 0 \cdot 0.2 + 1 \cdot 0.5 + 4 \cdot 0.3$

$= \sum y P_{y_i} =$ ↑

Continuous

$Z = X^3$

$E[Z] = \int z P_z\, dz$

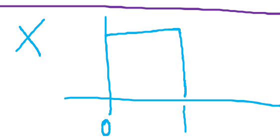

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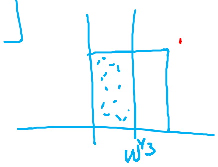
Illustration (Continuous)

$P_Z = P_X \left|\frac{dx}{dz}\right|$ $\quad Z = X^3$ $\quad \int z P_Z \, dz$

$\int g(x) P_X \, dx$

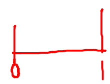
$F_Z(w) = P[Z \le w] = P[X^3 \le w] = P[X \le w^{1/3}]$

$= w^{1/3}$



$pdf_Z(w) = F'(w)$

$= \frac{1}{3} w^{-2/3}$



$\therefore E[Z] = \int w \frac{1}{3} w^{-2/3} \, dw = \frac{1}{4}$

$E[Z] = \int z \, pdf_X \, dx$

$= \int x^3 \, dx$

$= 1/4$

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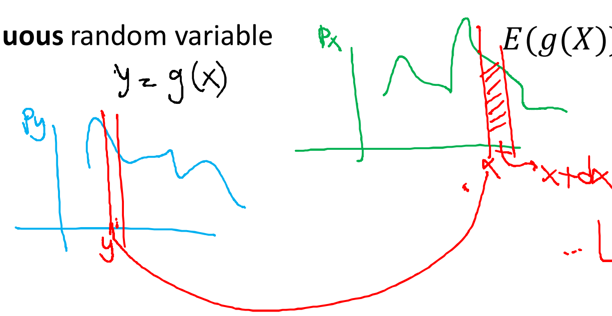
Function of a random variable

$Y = g(X)$ $\quad E[Y] = \sum y P_y$

For a function *g*(*X*) of a **discrete** random variable $E(g(X)) = \sum_i g(x_i) P(X = x_i)$

For a **continuous** random variable $E(g(X)) = \int_{-\infty}^{+\infty} g(x) f_X(x) \, dx$

$y = g(x)$



$P_x \, dx = P_y \, dy$

$\therefore$ Leibnitz $P_y = P_x \left|\frac{dx}{dy}\right|$

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Expected value

Kalshi

- Roulette: A ball settles in one of 38 numbered pockets
  - Guess the right number and you win INR 35 (probability 1/38);
  - Otherwise you lose INR 1 (probability 37/38).
- Expected earnings per trial
  - $(-1)(37/38) + (35)(1/38) = -0.0526$
- The house edge is exactly this negative expectation.

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Expected Value May Be Technically Undefined

$$P(X = x) = k/x^2 \text{ for } x \geq 1,\ x \in \mathbb{Z}^+$$

$$\sum_{x=1}^{\infty} P\,(X = x) = \sum_{x=1}^{\infty} k\,/x^2 = 1 \text{ if } k = 6/\pi^2$$

$$E(X) = \sum_{x=1}^{\infty} x\,P(X = x) = k \sum_{x=1}^{\infty} 1\,/x = \infty$$

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